

Some formulas in Queueing Theory

$M/G/1$ queue

$$P_L(z) = \frac{(1-\rho)\tilde{B}(\lambda-\lambda z)(1-z)}{\tilde{B}(\lambda-\lambda z)-z}$$

$$\tilde{W}(s) = \frac{(1-\rho)s}{s-\lambda+\lambda\tilde{B}(s)}$$

$$\tilde{R}(s) = \frac{1-\tilde{B}(s)}{sEB}$$

$$\tilde{B}P(s) = \tilde{B}(s+\lambda-\lambda\tilde{B}P(s))$$

$G/M/1$ queue

$$\sigma = \tilde{A}(\mu - \mu\sigma)$$

$M/G/1$ queue, NP-priorities

$$EW_i = \frac{\sum_{j=1}^r \rho_j ER_j}{(1 - \sum_{j=1}^i \rho_j)(1 - \sum_{j=1}^{i-1} \rho_j)}, \quad i = 1, \dots, r$$

$M/G/1$ queue, group arrivals

$$EW = \frac{\rho}{1-\rho}ER + \frac{(EG^2 - EG)EB}{2EG(1-\rho)}$$

$M/G/c/c$ queue

$$B(c, \rho) = \frac{\rho^c/c!}{\sum_{n=0}^c \rho^n/n!} = \frac{\rho B(c-1, \rho)}{c + \rho B(c-1, \rho)}, \quad (\text{with } \rho = \lambda EB)$$

$M/M/c$ queue

$$\Pi_W = \frac{(c\rho)^c}{c!} \left((1-\rho) \sum_{n=0}^{c-1} \frac{(c\rho)^n}{n!} + \frac{(c\rho)^c}{c!} \right)^{-1} = \frac{\rho B(c-1, c\rho)}{1-\rho + \rho B(c-1, c\rho)},$$

$$(\text{with } \rho = \frac{\lambda}{c\mu})$$